

(everyone can see) and private (secret and only for the owner). The two keys are connected mathematically in that the private key is used to derive the public key. With current technology, it is not feasible to derive the private key from the public key (hence, the description “asymmetric”). Users can receive a payment to their public address and spend it with their private key. Also see *symmetric key cryptography*.

Atomic. A provision that causes contract terms to revert as if tokens never left the starting point, if any contract condition is not met. An important feature of a *smart contract*.

Automated market maker (AMM). A *smart contract* that holds assets on both sides of a trading pair and continuously quotes a price for buying and for selling. Based on executed purchases and sales, the contract updates the asset size behind both the bid and the ask and uses this ratio to define a pricing function.

Barter. A peer-to-peer exchange mechanism in which two parties are exactly matched. For example, A has two pigs and needs a cow. B has a cow and needs two pigs. There is some debate as to whether barter was the first method of exchange. For example, David Graeber argues that the earliest form of trade was in the form of debit–credit. People living in the same village gave each other “gifts,” which by social consensus had to be returned in future by another gift that is usually a little more valuable (interest). People kept track of exchanges in their minds as it was only natural and convenient to do so since there is only a handful sharing